Algorithm 1 Meta-evolution with reevaluation

Inputs: baseline; G generations; O offspring per generation; P population size; tasks T; N_{init} initial seeds; B reevaluation seeds per generation; N_{val} validation seeds; F fitness metric.

`population` $\leftarrow$ [`baseline`]; `archive` $\leftarrow \{\}$
for $g = 1, \ldots, G$ **do**
  for $o = 1, \ldots, O$ **do** # Sample offspring
    `parents` $\leftarrow$ SamplePARENTS(`population`)
    $x_o \leftarrow$ LLM(`parents`); EVALUATE(x_o, T, N_{init})
    `archive` $\leftarrow$ `archive` $\cup \{x_o\}$
  end for
  # Reevaluate: spend B extra seeds across the archive
  $\hat{\mu}_x, n_x \leftarrow$ mean score and seed count of each $x \in$ `archive`
  if `strategy` = `uniform` **then**
    Split B seeds evenly over the top-P members of `archive` by $\hat{\mu}$
  else if `strategy` = `TTTS` **then** # Top-two Thompson sampling
    $\psi_x \leftarrow$ top-two Thompson probability of x under posterior $\mathcal{N}(\hat{\mu}_x, \hat{\sigma}^2/n_x)$
    Draw B members from `archive` with probability $\propto \psi_x$
  end if
  Evaluate each drawn member on the new seeds; update $\hat{\mu}_x, n_x$
  # MAP-Elites population: one niche per task
  `population` $\leftarrow \{\arg\max_{x \in \texttt{archive}} \hat{\mu}_x(t) \mid t \in T\}$
  Fill remaining slots (up to P) by highest overall $\hat{\mu}_x$
end for
# Final reevaluation on validation tasks
for $x \in$ `population` **do**
  `val`$[x] \leftarrow$ EVALUATE($x, T_{\text{val}}, N_{\text{val}}$)
end for
return $\arg\max_{x \in \texttt{population}}$ `val`$[x]$